

EOX PFR

Example: The liquid phase reaction $2A \rightarrow B$ is being carried out isothermally in a PFR. The rate law is kC_A^2 and the rate constant equals 0.03 L/mol*s. Species A is fed to the reactor at a concentration of 2 M and a flowrate of 3 L/s. Species A exits the reactor at a concentration of 0.1 M. Determine:

- The volume of the PFR.
- The rate of reaction in the PFR

Decision Guide: Isothermal PFR Design Example

Goal

Determine:

- The volume of the PFR.
- The reaction rate at the reactor exit.

Step 1: Identify the Type of Problem

Ask yourself:

What reactor?

→ PFR

Isothermal?

→ Yes

Liquid phase?

→ Yes

Because the reaction occurs in the liquid phase and the reactor is isothermal:

Volumetric flowrate is constant.

This greatly simplifies the concentration relationships.

Step 2: List the Known Information

The problem provides:

Reaction: $2A \rightarrow B$

Rate law: $-r_A = kC_A^2$

$k = 0.03 \text{ L/mol}\cdot\text{s}$

$C_{A0} = 2 \text{ M}$

$v_0 = 3 \text{ L/s}$

CA = 0.1 M at the reactor exit

Step 3: Calculate the Feed Molar Flowrate

Use:

$$F_{A0} = C_{A0}v_0$$

to determine the inlet molar flowrate of A.

This value will appear in the PFR mole balance. Because the reaction occurs in the liquid phase at constant volumetric flowrate:

$$F_A = C_A v_0$$

throughout the reactor.

Step 4: Choose the Appropriate Design Equation

Because the problem asks for reactor volume, begin with the PFR mole balance:

$$\frac{dF_A}{dV} = r_A$$

CBE 3610 Reactor Design Handbook Equation (16).

Since the volumetric flowrate is constant,

$$F_A = C_A v_0$$

and therefore

$$\frac{dF_A}{dV} = v_0 \frac{dC_A}{dV}.$$

Substituting into the mole balance gives:

$$v_0 \frac{dC_A}{dV} = r_A.$$

This form allows us to solve directly in terms of concentration.

Step 5: Substitute the Rate Law into the Mole Balance

The rate law is:

$$-r_A = kC_A^2.$$

Substitute into the PFR mole balance:

$$v_0 \frac{dC_A}{dV} = -kC_A^2$$

and rearrange into an integrable form:

$$dV = -\frac{v_0}{kC_A^2} dC_A.$$

Integrating from the inlet concentration to the exit concentration gives:

$$V = -\int_{C_{A0}}^{C_A} \frac{v_0}{kC_A^2} dC_A.$$

This is the PFR design equation written entirely in terms of concentration.

Step 6: Integrate to Obtain the Reactor Volume

Integrate from:

$$C_A = 2.0 \text{ M}$$

to

$$C_A = 0.10 \text{ M}$$

to determine the required reactor volume.

Step 7: Calculate the Exit Reaction Rate

The problem also asks for the reaction rate at the reactor exit. Use the exit concentration:

$$C_A = 0.10 \text{ mol/L}$$

and substitute directly into the rate law:

$$r_A = -(0.03)(0.10)^2$$
$$-r_A = kC_A^2$$

which gives the reaction rate at the reactor outlet.

Concept Check

Known Exit Concentration



Apply PFR Design Equation



Integrate for Volume



Use Exit Concentration Again



Calculate Exit Reaction Rate to obtain the reaction rate at the reactor exit.

Final Answer Checklist

- ☒ Calculated the inlet molar flowrate.
- ☒ Applied the PFR mole balance.
- ☒ Integrated to determine reactor volume.
- ☒ Calculated the reaction rate at the reactor exit.

CBE 3610 Reactor Design Handbook Equations Needed

Eq. (16) PFR Mole Balance

Eq. (44) General Rate Law

Eq. (45) Power-Law Kinetics

Key Insight

Concentration



Determines Rate



Rate Determines Required Reactor Volume

For second-order reactions, the reaction rate decreases rapidly as the reactant concentration falls. As a result, achieving the last portion of conversion often requires a disproportionately large amount of reactor volume.